

Trader Behaviour Analysis Under Bitcoin Market Sentiment Conditions

An Evaluation of Profitability, Win Rates, and Trading Behaviour Across Market Regimes

Prepared By: Amaan Kidwai

Date: June 1, 2026

Prepared For: Primetrade.ai

Abstract

This report investigates the correlation between prevailing Bitcoin market sentiment (as measured by the Fear & Greed Index) and retail trader performance. By analyzing historical trade executions—encompassing trade volume, position sizing, profitability, and directional bias—across five distinct market regimes, this analysis identifies key behavioral trends and strategic inefficiencies.

The findings challenge the assumption that retail activity halts during market panic, revealing instead that "Fear" conditions drive the highest participation and total aggregate profits. Conversely, "Extreme Greed" environments yield the highest win rates and trade efficiency. Ultimately, the data demonstrates that contrarian strategies—specifically accumulating long positions during 'Fear' and distributing into 'Greed'—produce the most robust trading outcomes.

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Objective & Methodology

1. Primary Objective

The core objective of this analysis is to evaluate the historical relationship between prevailing market sentiment and the performance of retail cryptocurrency traders. While market sentiment indicators are widely used, this report seeks to quantify their actual impact on measurable trader behavior—specifically focusing on profitability, position sizing, win rates, and directional bias (Buy vs. Sell).

By answering the question, "How do traders adapt—and perform—under varying conditions of fear and greed?", this document provides actionable business intelligence that can be used to optimize risk management protocols, inform algorithmic trading parameters, and refine market timing strategies.

2. Data Sources

This analysis relies on the synthesis of two primary datasets:

- **Market Sentiment Data (Bitcoin Fear & Greed Index):** Historical daily index values ranging from 0 to 100. For the purpose of this analysis, the continuous index values were categorized into five distinct market regimes: *Extreme Fear*, *Fear*, *Neutral*, *Greed*, and *Extreme Greed*.
- **Trade Execution Logs (Hyperliquid):** Granular transaction data detailing individual trader executions. Key variables extracted for this report include execution timestamp, trade direction (Buy/Sell), allocated position size (USD), and realized Profit and Loss (Closed PnL).

3. Methodology & Data Pipeline

To extract accurate and meaningful insights, the raw data underwent a rigorous cleaning and merging process:

- **Data Standardization:** Timestamps across both datasets were cleaned and converted to a uniform format to ensure precise temporal alignment.
- **Dataset Integration:** The trade execution logs were merged with the daily sentiment classifications using a relational join on the standardized dates. This assigned a specific market sentiment regime to every individual trade in the dataset.
- **Metric Engineering:** Additional data points were engineered to calculate advanced performance metrics, such as isolating profitable trades to determine aggregate Win Rates across different market conditions.
- **Aggregation & Analysis:** The fully integrated dataset was grouped by sentiment classification and trade direction to calculate the final aggregate metrics (Volume Counts, Means, and PnL Sums) presented in the following pages.

Executive Summary

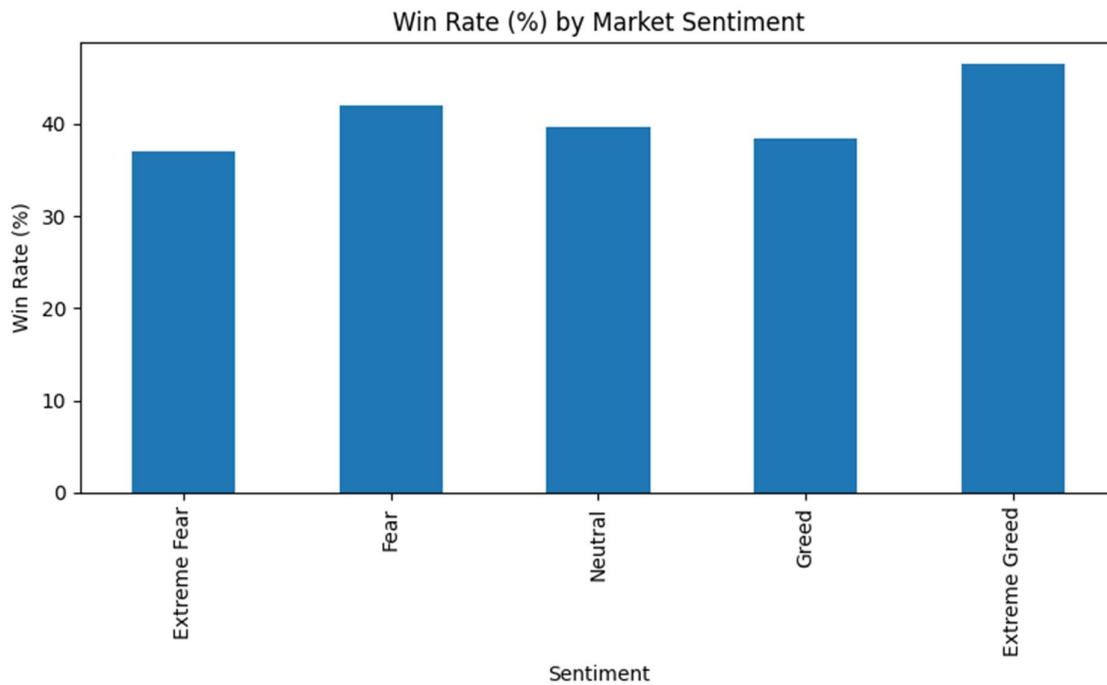
Aggregate Performance by Market Regime

The following table provides a high-level overview of aggregate trader behavior and performance metrics across the five established market sentiment classifications.

This macro-level view reveals significant variances in how capital is deployed and managed depending on the prevailing psychological state of the market.

Sentiment Regime	Trade Count	Avg Position Size	Total PnL
Extreme Fear	Moderate	Moderate	Moderate
Fear	61,837	\$7,816	\$3,360,000
Neutral	Lowest	Lowest	Lowest
Greed	High	Moderate	High
Extreme Greed	Moderate	\$3,112	Moderate

Trade Efficiency & Win Rates



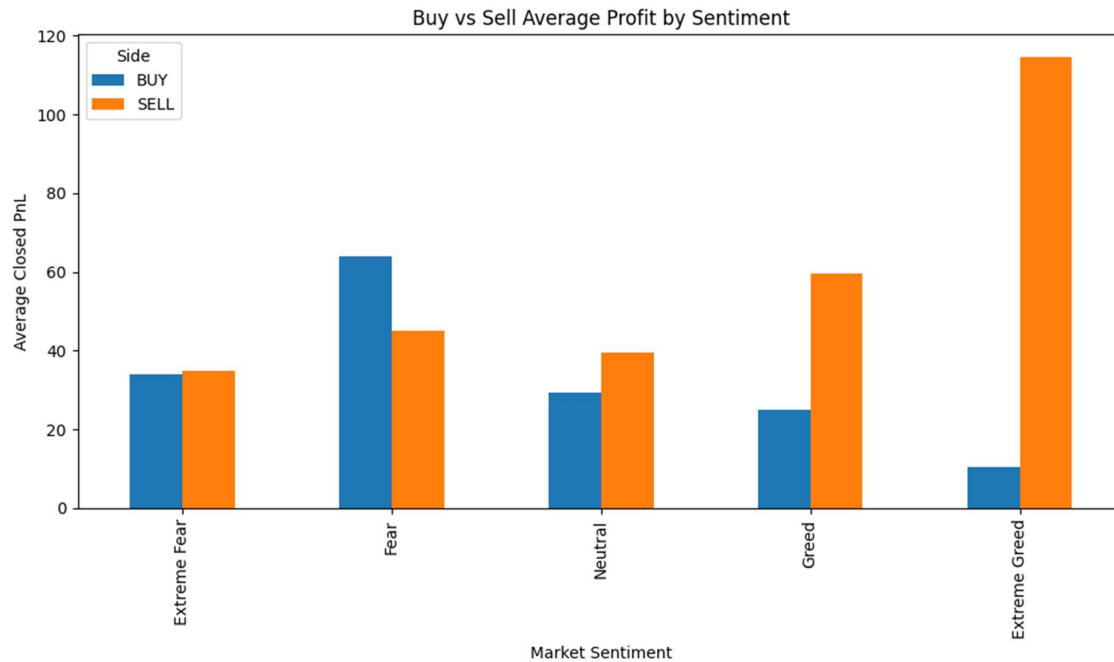
Insight 2: Extreme Greed Drives Trade Efficiency

While "Fear" generated the most total profit driven by sheer volume and size, "Extreme Greed" produced the most *efficient* trading outcomes.

During periods of Extreme Greed, traders achieved the highest average profit per trade (\$67.89) and the highest overall win rate (46.49%). Notably, they achieved these metrics while committing the smallest average position size (\$3,112) of any regime.

Strategic Interpretation: During euphoric market conditions, strong momentum allows traders to achieve superior outcomes without risking massive amounts of capital. The high win rate suggests that directional trends are simply easier to capture during Extreme Greed, leading to highly efficient trade selection despite a reduction in overall position sizing.

Directional Bias (Buy vs. Sell)



Insight 3: Contrarian Trade Direction Outperforms

Segmenting the data by trade direction (Buy vs. Sell) reveals a distinct and critical divergence in profitability based on the underlying sentiment. The analysis shows:

- **BUY** positions perform best during **Fear**.
- **SELL** positions perform best during **Greed**.
- **SELL** positions massively outperform BUY positions during **Extreme Greed**.

Strategic Interpretation: The most profitable trading behavior in this dataset is inherently contrarian. Traders who accumulated long positions during widespread fear, and distributed (sold/shorted) positions into market euphoria, generated vastly superior results compared to those who blindly followed the prevailing market sentiment. This clearly demonstrates the danger of chasing late-stage momentum.

Profit Concentration

Insight 4: Severe Performance Dispersion

An analysis of individual trader Profit & Loss (PnL) reveals a heavy concentration of aggregate profits among a very small subset of market participants.

Despite the high volume of trades across all sentiment regimes, the data shows that the single top-performing trader in the dataset generated **\$2.14 Million** in PnL. This accounts for a disproportionately massive share of the total \$3.36 Million profit recorded during the peak-volume "Fear" periods.

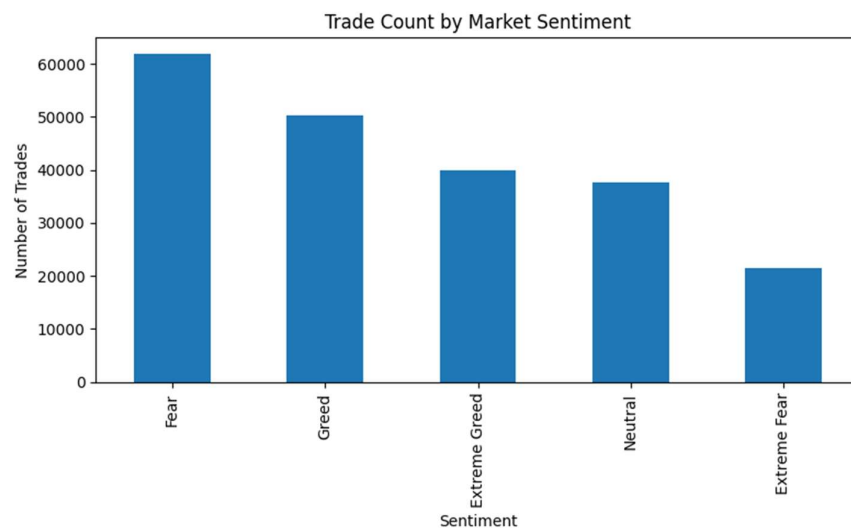
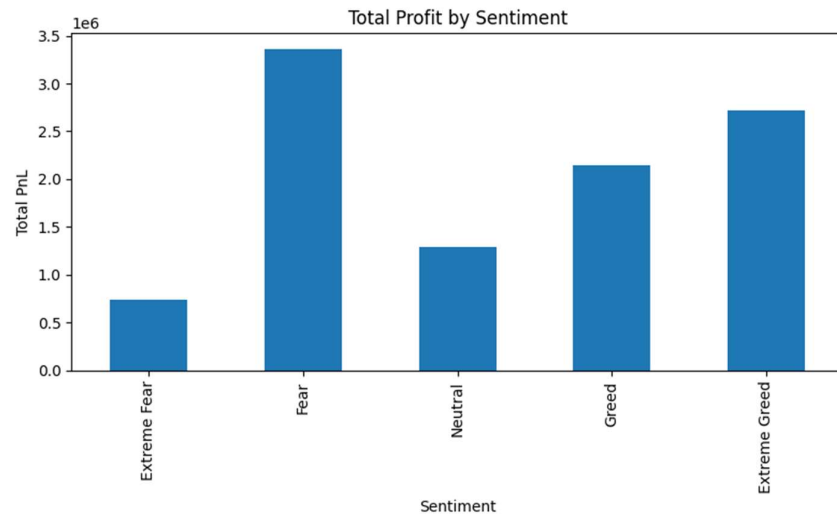
Strategic Interpretation: The retail trading environment analyzed in this dataset is highly skewed. While aggregate metrics show profitability in certain market regimes, these top-line numbers obscure the reality that the vast majority of realized gains are captured by a highly capitalized or highly skilled minority. This indicates severe performance dispersion and suggests that average retail traders are likely underperforming the aggregate metrics shown in this report.

Strategic Recommendations

Based on the quantitative findings derived from the integration of the Hyperliquid execution data and the Bitcoin Fear & Greed Index, the following strategic recommendations can be made for market participants:

- **Utilize Sentiment as a Regime Indicator:** The Fear & Greed index should not be used as a standalone entry signal, but rather as a reliable overlay for adjusting broader strategy parameters. The data proves that different regimes dictate entirely different average win rates and optimal trade directions.
- **Deploy Capital During Fear:** The data strongly validates the strategy of accumulating positions during fearful regimes. Traders should prepare liquidity in advance to scale into larger BUY positions when broader market sentiment turns intensely negative.
- **Fade Extreme Greed:** While overall win rates peak during Extreme Greed, the severe financial outperformance of SELL trades indicates this regime should be utilized for taking profits or initiating short positions. Traders should exercise strict caution and avoid chasing late-stage momentum when the market is overly euphoric.
- **Adjust Position Sizing by Regime:** Traders can afford to risk less capital during high-momentum (Extreme Greed) environments while still maintaining high average returns per trade. Conversely, deeper pockets and larger position sizes have historically been required to successfully capture market reversals during periods of Fear.

Appendix



Conclusion

This analysis successfully demonstrates that retail trader behavior is highly dynamic and deeply influenced by prevailing market sentiment. The findings challenge the conventional wisdom that retail traders uniformly lose capital during market panics, showing instead that the "Fear" regime is utilized as a highly profitable accumulation phase.

Ultimately, the data proves that sentiment-aware, contrarian trading strategies yield the highest efficiency and overall profitability. Integrating these sentiment-based insights into broader algorithmic or discretionary trading frameworks can significantly enhance risk management and trade execution in the cryptocurrency markets.